

DATE: ___/___/___

Quiz #4, Section G.

$$f(x) = \begin{cases} cx(5-x) & 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

a) value of c

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^5 cx(5-x) dx = 1$$

$$c \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^5 = 1$$

$$c \left[\left(\frac{125}{2} - \frac{125}{3} \right) - 0 \right] = 1$$

$$c \left(\frac{125}{6} \right) = 1$$

$$\boxed{c = 6/125}$$

CDF

$$b) F(x) = \int_{-\infty}^x f(x) dx = \int_0^x \frac{6}{125} x(5-x) dx$$

$$= \frac{6}{125} \int_0^x (5x - x^2) dx = \frac{6}{125} \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^x = \frac{3}{25} x^2 - \frac{2x^3}{125}$$

$$F(x) = \begin{cases} 0 & , x < 0 \\ -\frac{2x^3}{125} + \frac{3}{25}x^2 & , 0 \leq x \leq 5 \\ 0 & , x > 5 \end{cases}$$

$$c) P(1 < x < 3) = \overset{\text{cdf}}{F(3) - F(1)}$$

$$= \left[-\frac{2(3)^3}{125} + \frac{3}{25}(3)^2 \right] - \left[-\frac{2}{125} + \frac{3}{25} \right]$$

$$= \boxed{0.544}$$

$$\overset{\text{pdf}}{\int_1^3} \frac{6}{125} (5x - x^2) dx = \frac{6}{125} \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_1^3 = \boxed{0.544}$$

$$\begin{aligned}
 d) \quad E(x) &= \int_{-\infty}^{+\infty} x f(x) dx \\
 &= 6 \int_0^5 x (5x - x^2) dx \\
 &= \frac{6}{125} \left| \frac{5x^3}{3} - \frac{x^4}{4} \right|_0^5
 \end{aligned}$$

$$E(x) = 2.5$$

$$\begin{aligned}
 e) \quad V(x) &= ? \\
 E(x^2) &= \int_{-\infty}^{+\infty} x^2 f(x) dx \\
 &= 6 \int_0^5 x^2 (5x - x^2) dx \\
 &= 6 \int_0^5 (5x^3 - x^4) dx \\
 &= \frac{6}{125} \left| \frac{5x^4}{4} - \frac{x^5}{5} \right|_0^5
 \end{aligned}$$

$$E(x^2) = 7.5$$

$$\begin{aligned}
 V(x) &= E(x^2) - (E(x))^2 \\
 &= 7.5 - (2.5)^2 = 1.25
 \end{aligned}$$